

Rooftop solar is also a better option than the home inverters used today. The cost of power from home inverters is twice the cost of power from the utility. Combined efficiency of most of the branded home inverter-charger combinations is below 50 per cent. For 1kWh energy from them, home inverters cost the user 2 kWh of energy from the grid. In the case of solar plants, on the other hand, energy is free. Solar energy cost is simply related to the depreciation cost of the plant, as Sun gives its energy absolutely free.

Challenges to growth in this market

"The most important problem associated with home solar power is the customer's ignorance of the solar plant. There can be a scope today for a sweet-tongued vendor to sell unworthy sub-standard product. Government inspectors, in most cases, share profits with the supplier, so the buyer can't depend on them. Corruption is rampant wherever there is a question of subsidy. What is the value of a 20-year guarantee for the PV panels given by all sorts of 'solar manufacturers,' who have come up recently like mushrooms all over India," says Deodhar.

"Customers should buy from professional companies and not fall for low prices of non-reputed vendors. After all, buying a solar system makes sense only if it is durable and is a result of a good design and careful manufacturing," he suggests.

There are also some technical aspects about the rooftop solar systems (terrace tops in urban towns) that need attention.

"Rooftop solar systems permitted today are off-grid. Solar power that is mainly used in day time, like at all commercial and industrial locations, needs a small-storage-capacity battery to take care of small night load/evening load portion of the load. The system has to be designed with this aspect to contain the cost. Some states allow such systems to feed the en-

Tips for buyers

Buyers should ensure that all the components of the home solar system are provided by the supplier.

Use of sealed maintenance-free (SMF) batteries is recommended. When protected against deep discharge and over-charge, their life will be minimum three to four years since there is only one charge/discharge each day. One need not apply heavy load on them, so there is no need to use flooded batteries and hence no associated maintenance cost. A well-designed home solar system needs no maintenance, except cleaning of the dust from the panels, the same way as one will clean his car or bike.

Ensure that PV modules are from a reputed source and MNRE-approved. Energy meter will allow the buyer to measure the energy output in a day. A 1000-watt-peak solar plant should give a minimum 3.5 kWh (units) of electrical energy each sunny day. One should get 3.5 units of energy per kWp rating of the system.

Ensure that the power inverter's claimed efficiency is 85 per cent minimum. Inverters from a good company are as efficient as 95 per cent or more, but these are expensive. Brand of the inverter is also important. All sorts of cheap Chinese inverter products are being imported. So be cautious.

Mounting structure is very important. Ensure that it is of zinc-plated mild steel rolled sections which are powder coated. Also ensure that it is structurally well designed to withstand wind speed up to 150 km/hour minimum.

— P. S. Deodhar, founder, MITRAMAX Energy

ergy output to power grid," explains Deodhar.

"There is a problem associated with grid-tied inverters. They stop functioning when the grid fails. So in India, till we find an inverter that keeps on working to feed the local load when the grid supply fails, grid-tied option is invalid. However, grid-tied option that feeds the load is ideal since the unused portion of the solar energy of the solar plant can be then 'sold' to utility, as is done in Europe," Deodhar adds.

According to Goel, challenges to growth in rooftop solar PV market include:

1. Need for dedicated and experienced stakeholders for delivery of quality products as well as optimal solutions
2. Availability and use of good-quality components in a system to ensure optimal performance over the entire expected lifetime of the product/project
3. Creating awareness amongst the users to adopt systems based on lifetime ownership cost and not just initial capital cost of the system/project
4. Proactive and inclusive policies from the government right to the local

administration to evolve an effective and efficient delivery mechanism to the doorstep of the customers

5. Rating by approved bodies to encourage good practices and procedures for all I&C activities, post-installation monitoring and analysis of systems' performance data

6. Third-party audit of installations to verify compliance to standard procedures and good practices

Technologies and equipment

"With the availability of feature-rich and more efficient equipment, it is now relatively easy to harness solar power and monitor it remotely from the comfort of one's desktop. Solar power now has the potential to generate commercial profits through electricity generated from the rooftop installation and fed into the local grid," informs Goel.

Basically, there are eight components of a home solar system: PV solar modules, maximum power point tracking (MPPT) control that ensures harvesting of maximum energy from the PV panels, its weather-protected mounting structure, weather-proof junction boxes, battery bank, battery bank's electronic